

Notes: Li (RFS, 2011)

Constraints, R&D Investment, and Stock Returns

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1 Big picture

- **Question.** Why is the empirical relation between financial constraints and average stock returns so mixed in the literature, while the relation between R&D intensity and returns is often positive? Can the two facts be explained together?
- **Core idea.** R&D is much less flexible than ordinary capital investment. If a financially constrained firm cannot raise the required funding at a key stage, it may have to suspend the R&D project altogether. That suspension destroys value and raises systematic risk.
- **Main theoretical mechanism.** The paper adds external financing constraints to the Berk–Green–Naik multi-stage R&D venture model. A firm’s risk premium rises when financing ability falls, and this effect is stronger when required R&D spending is high.
- **Main prediction 1.** The financial-constraints–return relation should be strongest among R&D-intensive firms. In low-R&D firms, the relation can be weak or flat because ordinary investment is easier to scale back.
- **Main prediction 2.** The R&D–return relation should be strongest among financially constrained firms. R&D is risky primarily when firms are close to suspension because they cannot finance continuation.
- **Main empirical result.** In Fama–MacBeth regressions, the interaction term between financial constraints and R&D intensity is robustly positive. In portfolio sorts, constrained-minus-unconstrained return spreads are large mainly inside high-R&D firms, while high-minus-low R&D spreads are large mainly inside constrained firms.
- **Headline magnitudes.** Using the Kaplan–Zingales index, the high-minus-low constraint spread among high-RDCA firms is about 0.60% per month in raw returns and 0.47% in characteristic-adjusted returns, but it is close to zero among low-RDCA firms. Using the WW index, the high-minus-low constraint spread among high-RDME firms is 0.82% raw and 0.87% characteristic-adjusted, but negative and insignificant among low-RDME firms.
- **Why the paper matters.** The paper links a real-investment margin – project continuation versus suspension – to the cross-section of expected returns. It also offers a way to reconcile two literatures that otherwise look disconnected: financing constraints and the pricing of intangible investment.
- **Limitation to remember.** This is not a quasi-experimental causal design. The empirical exercise is a theory-guided asset-pricing test based on multiple proxies, portfolio sorts, and robustness checks.

2 Data and measurement (key objects)

2.1 Sample construction

- **Data sources.** Compustat provides accounting variables and CRSP provides stock returns.
- **Universe.** Domestic common shares traded on NYSE, AMEX, and NASDAQ with available accounting and return data.

- **Period.** 1975–2007. The starting date is driven by FASB Statement No. 2, which standardized R&D expense reporting in 1975.
- **R&D screen.** The sample includes only firm-years with non-missing R&D expenditures.
- **Distress screen.** Observations with negative *real* sales growth are removed, following the financial-constraints literature, to reduce the confounding effect of distress and decline.
- **Average sample size.** The retained sample contains about 1,333 firms per year on average.
- **Market-cap coverage.** Even after the R&D and sales-growth screens, the sample still covers about 43% of total U.S. market capitalization.

2.2 Descriptive sample facts

- **Deleted firms.** Non-R&D firms and R&D firms with negative real sales growth account for about 3,287 firms per year, with mean market capitalization around \$0.94 billion and median around \$0.10 billion.
- **Retained firms.** R&D-reporting firms with positive real sales growth are larger on average, with mean market capitalization around \$1.74 billion and median around \$0.18 billion.
- **Within R&D firms.** High-R&D firms tend to be smaller in median size than low-R&D firms, which is exactly why the paper later checks carefully whether the results are merely a disguised size effect.

2.3 Financial-constraints proxies

The paper uses five proxies.

$$KZ = -1.0019 \frac{\text{CashFlow}}{K} + 0.2826Q + 3.1392 \frac{\text{Debt}}{\text{Total Capital}} - 39.3678 \frac{\text{Dividends}}{K} - 1.3148 \frac{\text{Cash}}{K}. \quad (1)$$

$$SA = -0.737 \text{Size} + 0.043 \text{Size}^2 - 0.040 \text{Age}. \quad (2)$$

$$WW = -0.091 CF - 0.062 DIVPOS + 0.021 TLTD - 0.044 LNTA + 0.102 ISG - 0.035 SG. \quad (3)$$

- **Interpretation.** Higher KZ, higher WW, and higher SA all correspond to tighter financial constraints.
- **Additional proxies.** Younger age and smaller market capitalization are also treated as indicators of tighter financial constraints.
- **Why so many measures?** The paper wants the interaction result to survive measurement error in any single constraints proxy.
- **Empirical fact.** Size is highly correlated with WW and SA, but much less so with KZ. This matters for the triple-sort interpretation later.

2.4 R&D-intensity measures

Six measures of R&D intensity are used:

$$\text{RDA} = \frac{RD}{A}, \quad \text{RDCAP} = \frac{RD}{CAPX}, \quad \text{RDS} = \frac{RD}{Sales}, \quad \text{RDE} = \frac{RD}{Employees}, \quad \text{RDME} = \frac{RD}{ME}, \quad \text{RDC}$$

R&D capital is constructed as a depreciated stock of past R&D spending:

$$RDC_{it} = RD_{it} + 0.8 RD_{it-1} + 0.6 RD_{it-2} + 0.4 RD_{it-3} + 0.2 RD_{it-4}. \quad (4)$$

- **Interpretation.** RDCA captures a stock concept of accumulated intangible investment, while the other measures are flow concepts.
- **Why multiple scalings?** Scaling by assets, sales, employees, capital expenditures, or market equity checks that the result is not an artifact of any specific denominator.
- **Empirical fact.** The six R&D measures are highly rank-correlated, with Spearman correlations roughly between 0.70 and 0.96.

2.5 Return and control variables

- **Return horizon.** Monthly stock returns from July of year t to June of year $t + 1$.
- **Control variables.** The main Fama–MacBeth regressions control for $\ln(\text{ME})$, $\ln(\text{BE}/\text{ME})$, six-month momentum (with a one-month gap), and return on assets ROA.
- **Transformations.** In the regression tables, KZ, RDME, RDS, and $\ln(\text{ME})$ are expressed as demeaned percentiles, while other control variables are winsorized at the top and bottom 1%.
- **Adjusted returns in portfolio tests.** The paper reports raw value-weighted returns, DGTW characteristic-adjusted returns, and two-digit industry-adjusted returns.

3 Models and empirical specifications (all equations + interpretation)

3.1 Underlying cash flow and pricing kernel

The future cash flow of a completed R&D project follows a geometric Brownian motion:

$$dy(t) = \hat{\mu}y(t) dt + \sigma y(t) d\hat{w}(t). \quad (5)$$

The stochastic discount factor evolves as

$$dm(t) = -r m(t) dt + \theta m(t) d\hat{z}(t), \quad (6)$$

and the market price of risk for the cash flow process is

$$\lambda = \sigma\theta\rho. \quad (7)$$

Under the risk-neutral measure,

$$dy(t) = \mu y(t) dt + \sigma y(t) dw(t), \quad \mu = \hat{\mu} - \lambda. \quad (8)$$

- **Interpretation.** $\hat{\mu}$ is the physical drift of project cash flow; r is the risk-free rate; λ summarizes how much systematic risk the project's cash flow carries.
- **Economic role.** The higher is systematic exposure, the larger is the discount embedded in project value.

3.2 Completed-project value

Once the project successfully completes all N stages, the firm simply owns the cash-flow stream:

$$V(y(t), N(t)) = \frac{y(t)}{r - \mu}. \quad (9)$$

- **Interpretation.** This is the continuous-time Gordon–Williams formula.
- **Why it matters.** All of the paper's action comes *before* completion, when financing frictions can force suspension.

3.3 Bellman equation with financing constraints

Before completion, the firm chooses whether to continue investing ($\nu = 1$) or suspend ($\nu = 0$):

$$rV(y, n) = \max_{\nu \in \{0,1\}} \left\{ -\nu x(n) + \frac{1}{dt} \mathbb{E}_t[dV(y, n)] \right\}, \quad (1)$$

subject to

$$p(n) \frac{1}{dt} \mathbb{E}_t[dV(y, n)] \geq x(n). \quad (2)$$

- **Interpretation of $x(n)$.** $x(n)$ is the required investment at stage n . Crucially, it is *not* chosen by the firm. It is technologically or regulatorily required.
- **Interpretation of $p(n)$.** Only a fraction $p(n) \in (0, 1)$ of the expected increase in firm value from continuing can be pledged to outside financiers.
- **Economic intuition.** Hidden information, agency problems, and low pledgeability are summarized by $p(n)$. Lower $p(n)$ means tighter financing constraints.
- **Key asymmetry between R&D and ordinary investment.** Because $x(n)$ is externally imposed, the firm cannot smoothly scale down R&D when financing conditions worsen. It must often suspend.

3.4 Hamilton–Jacobi–Bellman equation

Applying Itô's lemma yields the HJB:

$$rV(y, n) = \frac{1}{2} \sigma^2 y^2 \frac{\partial^2 V(y, n)}{\partial y^2} + \mu y \frac{\partial V(y, n)}{\partial y} + \max_{\nu \in \{0,1\}} \nu \{ \pi(n) [V(y, n+1) - V(y, n)] - x(n) \}. \quad (3)$$

- **Interpretation of $\pi(n)$.** $\pi(n)$ is the success intensity at stage n , so the firm values the expected jump in continuation value from advancing to the next stage.
- **Inside the max.** The firm compares the expected jump in value from advancing to stage $n + 1$ against the required R&D outlay $x(n)$.
- **Project-level intuition.** A biotech firm finishing Phase II and moving to Phase III is the motivating example in the paper.

3.5 Threshold rules: continuation versus mothball

There are two separate thresholds:

- the **cost-benefit threshold** $y_{CB}^*(n)$, pinned down by whether investment is privately worthwhile; and
- the **financing threshold** $y_{FC}^*(n)$, pinned down by whether the firm can raise enough external funding.

The firm continues only if

$$y \geq y^*(n) = \max \{y_{CB}^*(n), y_{FC}^*(n)\}. \quad (10)$$

- **Interpretation.** Even if a project is positive-NPV, the firm may still suspend it when financing capacity is too low.
- **Economic message.** For constrained firms, financing conditions, not just fundamentals, determine project continuation.

3.6 Closed-form value function and boundary conditions

Appendix A writes firm value as a piecewise function:

$$V(y, n) = \begin{cases} V^m(y, n) = D(n)y^\beta, & y < y^*(n), \\ V^c(y, n) = \sum_{i=n}^{N-1} C(i, n)y^{\gamma(i)} + B(n)y + A(n), & y \geq y^*(n). \end{cases} \quad (A1)$$

The boundary conditions are

$$V^c(y^*(n), n) = V^m(y^*(n), n), \quad (A4)$$

$$\frac{\partial}{\partial y} V^c(y^*(n), n) = \frac{\partial}{\partial y} V^m(y^*(n), n), \quad (A5)$$

$$\pi(n)[V(y_{CB}^*(n), n+1) - V(y_{CB}^*(n), n)] = x(n), \quad (A6)$$

$$p(n)\frac{1}{dt}\mathbb{E}_t[dV(y_{FC}^*(n), n)] = x(n). \quad (A7)$$

The mothball-region exponent satisfies

$$\beta = \frac{(\sigma^2 - 2\mu) + \sqrt{8\sigma^2 r + (2\mu - \sigma^2)^2}}{2\sigma^2} > 1. \quad (A9)$$

- **Interpretation.** In the mothball region, the firm's value is essentially an option to restart later. In the continuation region, value combines the current project, future advancement opportunities, and the option to suspend.
- **Why the appendix matters.** These closed forms allow the paper to sign the derivatives of the risk premium with respect to financing ability and R&D intensity.

3.7 Risk premium formula

The instantaneous expected excess return is

$$R(y, n) = \frac{V_y(y, n) y}{V(y, n)} \lambda. \quad (4)$$

- **Interpretation.** The firm's risk premium is the cash-flow price of risk λ scaled by the elasticity of firm value with respect to the underlying project cash flow.
- **Key intuition.** When the firm is close to the suspension threshold, value becomes more option-like and more sensitive to systematic cash-flow shocks. That raises expected returns.

3.8 Comparative statics: financing ability and required R&D

For firms near the final stage and in the region where financial constraints bind more tightly than the pure cost-benefit condition, the paper proves:

$$\frac{\partial R(n)}{\partial p(n)} < 0, \quad \frac{\partial^2 R(n)}{\partial p(n) \partial x(n)} < 0, \quad (5-6)$$

and

$$\frac{\partial R(n)}{\partial x(n)} > 0, \quad \frac{\partial^2 R(n)}{\partial x(n) \partial p(n)} < 0. \quad (7-8)$$

- **Interpretation of Proposition 1.** Tighter financing ability (lower p) raises expected returns, but mainly when the firm is R&D-intensive enough that financing constraints are the binding continuation margin.
- **Interpretation of Proposition 2.** Higher required R&D spending raises expected returns, but mainly among firms with weak financing ability.
- **Empirical translation.** The slope on Constraints $\times$ R&D in a return regression should be positive.

3.9 Numerical illustrations used in the paper

The figures are built from a five-stage calibration with

$$\hat{\mu} = 3\% \text{ per year}, \quad \sigma = 40\% \text{ per year}, \quad r = 17.54\% \text{ per year}, \quad \lambda = 8\% \text{ per year},$$

$\pi(1) = 1$ rising by 0.1 per completed stage, financing ability $p \in [0.35, 0.8]$, and required R&D spending that rises by 3 per stage.

- **Interpretation.** The calibration is not meant to pin down magnitudes exactly. It is a graphical device showing that the comparative statics are stage-dependent and especially strong for high-R&D, low- p firms.

3.10 Baseline Fama–MacBeth return regression for financial constraints

To replicate Lamont, Polk, and Saa-Requejo (2001), the paper estimates

$$R = a + \gamma_1 \text{Constraints} + \gamma_2 \ln(\text{ME}) + \gamma_3 \ln(\text{BE/ME}) + \gamma_4 \text{Momentum} + \gamma_5 \text{ROA}. \quad (9)$$

- **Purpose.** This regression checks whether the paper can reproduce the standard mixed evidence on the constraints–return relation before adding the R&D interaction.
- **Result.** It can. The KZ slope is negative and insignificant in both the original LPS manufacturing sample and the paper’s extended sample.

3.11 Baseline Fama–MacBeth return regression for R&D

To replicate Chan, Lakonishok, and Sougiannis (2001), the paper estimates

$$R = a + \gamma_1 \text{R\&D} + \gamma_2 \ln(\text{ME}) + \gamma_3 \ln(\text{BE/ME}) + \gamma_4 \text{Momentum} + \gamma_5 \text{ROA}. \quad (10)$$

- **Purpose.** This regression recovers the familiar positive R&D–return relation.
- **Result.** It does. The slope on RDME and RDS is strongly positive in both the CLS sample and the extended sample.

3.12 Main interaction regression

The central empirical test is

$$R = a + \gamma_1 \text{Constraints} + \gamma_2 \text{R\&D} + \gamma_3 \text{Constraints} \times \text{R\&D} + \gamma_4 \text{ROA} + \gamma_5 \ln(\text{ME}) + \gamma_6 \ln(\text{BE/ME}) + \gamma_7 \text{Momentum}. \quad (11)$$

- **Core null.** $\gamma_3 = 0$: the pricing of constraints does not depend on R&D intensity.
- **Model prediction.** $\gamma_3 > 0$.
- **Main result.** The estimate of γ_3 is positive and significant at the 1% or 5% level for all six R&D measures when constraints are proxied by KZ, and similarly in unreported regressions for the other constraints measures.
- **Size check.** Adding a three-way term $\text{KZ} \times \text{R\&D} \times \ln(\text{ME})$ leaves the two-way interaction positive and significant, implying the effect is not driven solely by microcaps.

3.13 Portfolio-sort design

At the end of June in year t , firms are sorted independently into:

- two R&D groups,
- three constraints groups,
- and, in some tests, two size groups as well.

The intersections generate double-sort or triple-sort portfolios held for the next 12 months.

- **Why independent sorts?** This keeps the test focused on the interaction rather than mechanically embedding one characteristic into the other.
- **What the hedge portfolios mean.** A constrained-minus-unconstrained return spread within a given R&D bin tests whether constraints matter more when R&D is high. A high-minus-low R&D spread within a given constraints bin tests whether R&D matters more when firms are constrained.

4 Identification and empirical design (what makes the evidence credible)

4.1 This is not an IV or natural-experiment paper

- The paper does *not* identify causal effects through an external instrument, policy reform, or discontinuity.
- Instead, the empirical design is a theory test with a sharp interaction prediction: if R&D inflexibility is the real mechanism, then the constraints–return relation must strengthen with R&D intensity, and the R&D–return relation must strengthen with constraints.

4.2 What variation is doing the work

- Cross-sectional variation in financial-constraints proxies among R&D-reporting firms.
- Cross-sectional variation in R&D intensity, measured in six different ways.
- Monthly cross-sectional return variation, summarized either through Fama–MacBeth regressions or portfolio sorts.

4.3 Why the paper is more persuasive than a single proxy result

- **Multiple constraints measures.** KZ, WW, SA, age, and size all point in the same qualitative direction.
- **Multiple R&D measures.** The interaction survives scaling R&D by assets, sales, employees, capital expenditures, market equity, and an R&D-capital stock.
- **Distress control.** Dropping firms with negative real sales growth is important because otherwise a positive “constraints premium” might just reflect distress risk.
- **Size control.** Triple sorts and three-way regressions show the result is not just a relabeled small-firm effect.

- **Mechanism consistency.** The factor-loading results line up with the idea that constrained high-R&D firms bear more market and especially size-related systematic risk.

4.4 What the paper rules out

- **Pure leverage or distress story.** Sorting on leverage does not generate a comparable return pattern; high-R&D firms are not simply high-debt firms.
- **Generic investment irreversibility.** Sorting on capital-investment intensity produces a flat KZ–return relation, which sharpens the claim that R&D’s inflexibility is special.
- **Only tiny firms.** The strongest KZ and age results inside high R&D often appear among *large* firms, not among the very smallest firms.

5 Tables and figures

5.1 Figure 1: Financing ability and risk premium

Read:

- At every stage, the risk premium falls as financing ability p rises. Better access to external finance lowers the chance of suspension and therefore lowers risk.
- The slope is much steeper for high-R&D firms than for low-R&D firms. That is the model’s central interaction prediction in picture form.
- The gap between high and low R&D becomes smaller at later stages because the project is closer to completion, so the option to suspend matters less.

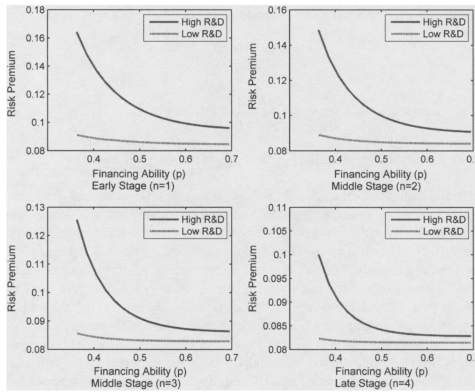


Figure 1
Financing ability and risk premium
The risk premiums (per annum) of projects that require five stages to complete are plotted as a function of their financing abilities (p) for different levels of R&D investment and over different stages. For example, the top left plot is for projects that have completed the first stage ($n = 1$), the top right for projects that have completed two stages ($n = 2$), and so forth. The risk premium is the instantaneous expected return minus the risk-free rate. The two lines correspond to different levels of R&D investment, which starts from 1 for low R&D firms and 10 for high R&D firms. The R&D investment increases by 3 with each additional completed stage. The volatility of future cash flow $\sigma = 0.4$ and the success intensity begin with $\pi(1) = 1$ and increase by 0.1 with each additional completed stage.

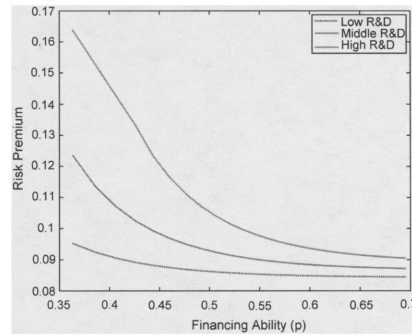


Figure 2
Financing ability and time-averaged risk premium
This figure plots the time-averaged risk premium against the financing ability (p) for firms with different levels of R&D investment (α). The project takes five stages to complete. The risk premium is averaged over the five stages. The three levels of R&D (α) start from 5, 10, and 15, respectively, and increase by 3 with each additional completed stage. The volatility of future cash flow $\sigma = 0.4$ and the success intensity begin with $\pi(1) = 1$ and increase by 0.1 with each additional completed stage.

5.2 Figure 2: Financing ability and time-averaged risk premium

Read:

- Averaging over the five stages preserves the same pattern: lower p means higher average risk premium.
- The high-R&D curve lies well above the low-R&D curve, and the negative financing-ability slope is clearly strongest for high R&D.
- This figure is the cleanest stage-averaged picture of why the paper expects a constraints premium mainly among R&D-intensive firms.

5.3 Figure 3: R&D investment and risk premium

Read:

- In the continuation region, the risk premium rises with required R&D spending x .
- The relation is much steeper for more constrained firms ($p = 0.3$) than for less constrained firms ($p = 0.7$).
- The flat portions correspond to the mothball region. Once the project is suspended, the relation between marginal R&D and risk behaves differently because the firm is now mostly an option to restart.

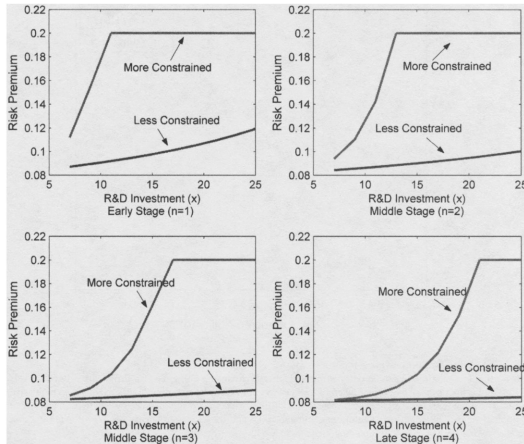


Figure 3
R&D investment and risk premium
The risk premiums (per annum) of projects that require five stages to complete are plotted as a function of their R&D investment requirement (x) for different financing abilities (p) and over different stages. For example, the top left plot is for projects that have completed the first stage ($n = 1$), the top right for projects that have completed two stages ($n = 2$), and so forth. The risk premium is the instantaneous expected return minus the risk-free rate. The two lines correspond to different levels of external financing abilities (p): 0.3 for more constrained firms and 0.7 for less constrained firms. The volatility of future cash flow $\sigma = 0.4$ and the success intensity begin with $\pi(1) = 1$ and increase by 0.1 with each additional completed stage.

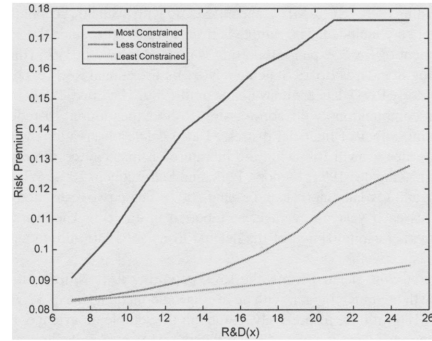


Figure 4
R&D investment and time-averaged risk premium
This figure plots the time-averaged risk premium against the R&D investment (x) for firms with different levels of financing ability (p). The project takes five stages to complete. The risk premium is averaged over the five stages. The three levels of financing ability (p) are 0.3, 0.5, and 0.7. The volatility of future cash flow $\sigma = 0.4$ and the success intensity begin with $\pi(1) = 1$ and increase by 0.1 with each additional completed stage.

5.4 Figure 4: R&D investment and time-averaged risk premium

Read:

- Averaging over stages again preserves the interaction: the R&D–risk relation is steepest for the most constrained firms.
- This is the graphical counterpart to the later finding that the positive R&D–return relation shows up only among financially constrained firms.

- The figure also helps interpret why some aggregate R&D-return evidence can look weak in the full sample: the slope is diluted by unconstrained firms.

5.5 Table 1: Summary statistics

Read:

- The retained R&D sample is not tiny in market-cap terms. It contains only 1,333 firms per year on average but still represents about 43% of the total U.S. stock market.
- High-R&D firms are generally smaller in median market capitalization than low-R&D firms. That motivates the later triple sorts by size.
- Relative to the dropped firms, the retained firms are larger, healthier, and more economically important. The paper is not just studying a fringe tail of very small speculative firms.

Table 1
Summary statistics

	No. of Firms	Average <i>ME</i> (\$bn)	Median <i>ME</i> (\$bn)	Total <i>ME</i> (\$bn)	<i>ME</i> (%)
Non-R&D firms and R&D firms with <i>RSGRO</i> ≤ 0	3287	0.94	0.10	3083	57
All R&D firms with <i>RSGRO</i> > 0	1333	1.74	0.18	2325	43
Low <i>RDA</i>	650	1.79	0.24	1166	22
High <i>RDA</i>	651	1.74	0.15	1130	21
Low <i>RDCA</i>	653	1.75	0.24	1142	21
High <i>RDCA</i>	654	1.78	0.15	1164	22
Low <i>RDCA</i>	641	2.00	0.26	1285	24
High <i>RDCA</i>	642	1.54	0.14	990	18
Low <i>RDS</i>	650	1.79	0.23	1161	21
High <i>RDS</i>	651	1.74	0.16	1135	21
Low <i>RDE</i>	640	1.51	0.22	966	18
High <i>RDE</i>	641	2.04	0.17	1310	24
Low <i>RDME</i>	645	2.33	0.27	1501	28
High <i>RDME</i>	645	1.22	0.13	789	15

At the end of June of year t , I sort R&D-reporting firms with positive real sales growth (*RSGRO*) into low and high R&D groups according to their R&D intensities in the fiscal year ending in year $t-1$. The portfolios are reformed every year. *RDA*, *RDCA*, *RDS*, *RDE*, and *RDME* are R&D expenditure scaled by assets, capital expenditure, sales, number of employees, and year-end market equity. *RDCA* is R&D capital scaled by assets, where R&D capital is the weighted sum of a firm's R&D expenditure over the past five years, assuming an annual amortization rate of 20%. *ME* is year-end market equity. I report the time-series mean of cross-sectional average number of firms, the mean and median market capitalization (in billions), total market capitalization, and the market capitalization percentage of all R&D firms and of low and high R&D firms. For comparison, I also report these summary statistics of non-R&D reporting firms and R&D-reporting firms with negative real sales growth in the top row. The sample period is from 1975 to 2007.

Table 2
Comparison of Fama-MacBeth regression results in different samples

Panel A. The KZ-return relation in LPS (2001) sample

<i>KZ</i>	$\ln(BE/ME)$	$\ln(ME)$	<i>Momentum</i>	<i>ROA</i>
-0.19 (-1.21)	0.44 (5.79)	-0.31 (-0.95)	0.88 (3.29)	0.75 (1.25)

Panel B. The KZ-return relation in the extended sample

<i>KZ</i>	$\ln(BE/ME)$	$\ln(ME)$	<i>Momentum</i>	<i>ROA</i>
-0.29 (-1.43)	0.36 (5.41)	-1.06 (-3.02)	0.70 (2.88)	0.22 (0.47)

Panel C. The R&D-return relation in Chan, Lakonishok, and Sougiannis (2001) sample

$\ln(BE/ME)$	$\ln(ME)$	<i>Momentum</i>	<i>RDME</i>	<i>ROA</i>	<i>RDS</i>
0.29 (3.85)	-1.12 (-3.02)	0.39 (1.79)	1.06 (5.30)	0.78 (1.83)	
0.34 (5.49)	-1.23 (-3.28)	0.40 (1.81)		0.69 (1.62)	0.80 (3.31)

Panel D. The R&D-return relation in the extended sample

$\ln(BE/ME)$	$\ln(ME)$	<i>Momentum</i>	<i>RDME</i>	<i>ROA</i>	<i>RDS</i>
0.31 (4.24)	-0.93 (-2.67)	0.67 (2.75)	1.27 (4.82)	0.48 (1.17)	
0.41 (6.83)	-1.03 (-2.91)	0.65 (2.74)		0.38 (0.93)	0.96 (2.85)

I report the time-series average slopes and their t -statistics (in parentheses) from Fama-MacBeth cross-sectional regressions within different samples. The Lamont, Polk, and Saa-Requejo (LPS, 2001) sample in Panel A includes manufacturing firms with positive real sales growth from 1968 to 1997. The extended sample in Panels B and D includes all firms with positive real sales growth from 1975 to 2007. The Chan, Lakonishok, and Sougiannis (2001) sample in Panel C includes all firms from 1975 to 1995. For each month from July of year t to June of year $t+1$, I run cross-sectional regressions of monthly percent returns on different variables: *KZ* is the Kaplan and Zingales (1997) index of financial constraints for the fiscal year ending in year $t-1$; $\ln(BE/ME)$ is the log book equity for the fiscal year ending in year $t-1$ minus the log market equity at the end of December of year $t-1$; $\ln(ME)$ is the log market capitalization at the end of June of year t ; *Momentum* is the prior six-month returns (with a one-month gap between the holding period and the current month); *ROA* is the net income scaled by total assets for the fiscal year ending in year $t-1$; *RDME* and *RDS* are R&D expenditure scaled by year-end market equity and sales, respectively, for the fiscal year ending in year $t-1$. *KZ*, *RDME*, *RDS*, and $\ln(ME)$ are demeaned percentiles. The other control variables are winsorized at the top and bottom 1%.

5.6 Table 2: Replicating the prior literatures

Read:

- Panel A and Panel B replicate the mixed constraints literature: the KZ slope is negative and statistically insignificant in both the Lamont–Polk–Saa-Requejo sample and the paper's extended sample.
- Panel C and Panel D replicate the positive R&D literature: the slope on RDME is 1.06 in the CLS sample and 1.27 in the extended sample, both strongly significant; the slope on RDS is also positive and significant.
- This table is important because it shows the paper is not overturning either earlier result. It is explaining *why both can coexist*.

Table 3
Slopes from Fama-MacBeth (1973) cross-sectional regressions of monthly percent returns on the KZ index, R&D intensity, KZ * R&D, and other control variables

Panel A: Without the interaction term $KZ * R\&D * \ln(ME)$								
	KZ	R&D	KZ* R&D	KZ* R&D* $\ln(ME)$	ROA	$\ln(ME)$	$\ln(BE/ME)$	Momentum
RDME	0.02 (0.09)	1.26 (5.15)	1.17 (2.71)		0.70 (1.62)	-0.93 (-2.68)	0.32 (4.60)	0.65 (2.72)
RDS	0.07 (0.46)	1.01 (3.06)	0.95 (2.40)		0.77 (1.83)	-1.03 (-2.91)	0.42 (7.19)	0.64 (2.72)
RDE	0.10 (0.61)	1.08 (3.40)	1.06 (2.67)		0.69 (1.61)	-1.10 (-3.10)	0.41 (6.87)	0.65 (2.77)
RDCA	0.10 (0.60)	1.19 (4.30)	0.78 (2.04)		0.69 (1.64)	-0.89 (-2.58)	0.42 (7.17)	0.65 (2.77)
RDCAP	0.08 (0.52)	1.02 (3.89)	1.15 (2.83)		0.73 (1.69)	0.94 (-2.68)	0.42 (6.90)	0.68 (2.84)
RDA	0.12 (0.78)	1.21 (4.02)	0.80 (1.96)		0.77 (1.83)	-0.96 (-2.76)	0.44 (7.42)	0.64 (2.71)

Panel B: With the interaction term $KZ * R\&D * \ln(ME)$								
	KZ	R&D	KZ* R&D	KZ* R&D* $\ln(ME)$	ROA	$\ln(ME)$	$\ln(BE/ME)$	Momentum
RDME	0.00 (-0.02)	1.22 (5.01)	1.03 (2.57)	-1.81 (-1.15)	0.74 (1.69)	-0.94 (-2.69)	0.32 (4.55)	0.66 (2.73)
RDS	0.07 (0.46)	0.96 (2.93)	0.90 (2.34)	-1.40 (-1.81)	0.78 (1.87)	-1.03 (-2.91)	0.42 (7.20)	0.65 (2.73)
RDE	0.09 (0.61)	1.04 (3.26)	1.04 (2.62)	-1.56 (-0.89)	0.70 (1.64)	-1.12 (-3.11)	0.41 (6.89)	0.65 (2.77)
RDCA	0.09 (0.58)	1.13 (4.08)	0.70 (1.92)	-2.90 (-1.90)	0.72 (1.70)	-0.93 (-2.68)	0.43 (7.20)	0.66 (2.78)
RDCAP	0.08 (0.47)	0.99 (3.77)	1.06 (2.74)	-1.06 (-0.74)	0.74 (1.70)	0.42 (-2.71)	0.42 (6.88)	0.69 (2.85)
RDA	0.12 (0.75)	1.15 (3.81)	0.73 (1.86)	-2.59 (-1.57)	0.79 (1.87)	-1.00 (-2.83)	0.44 (7.43)	0.65 (2.72)

In Panel A for each month from July of year t to June of year $t+1$ I run cross-sectional regressions of monthly percent returns on KZ , $R\&D$, and $KZ * R\&D$ measured in the fiscal year ending in year $t-1$, and other control variables: ROA is the net income scaled by total assets for the fiscal year ending in year $t-1$, $\ln(ME)$ is the natural log of market capitalization at the end of June of year t , $\ln(BE/ME)$ is the natural log of the ratio of book equity to market equity for the fiscal year ending in year $t-1$, and $Momentum$ is the prior six-month returns (with a one-month gap between the holding period and the current month). The first column shows measures of R&D intensity used in the regressions. RDA , $RDCAP$, RDS , RDE , and $RDME$ are defined as in Table 1. $RDCA$ is R&D capital scaled by assets, where R&D capital is the weighted sum of a firm's R&D expenditure over the past five years, assuming an annual amortization rate of 20%. In Panel B, an additional interaction term is included: $KZ * R\&D * \ln(ME)$. KZ , $RDME$, RDS , and $\ln(ME)$ are demeaned percentiles. The other control variables are winsorized at the top and bottom 1%. I report the slopes and their Fama-MacBeth t -statistics (in parentheses). The sample is from 1975 to 2007 and includes R&D-reporting firms with positive real sales growth only.

5.7 Table 3: Fama-MacBeth interaction regressions

Read:

- The slope on $KZ \times R\&D$ is positive and significant for all six R&D measures. For example, the interaction coefficient is 1.17 for $KZ \times RDME$, 0.95 for $KZ \times RDS$, 1.06 for $KZ \times RDE$, 0.78 for $KZ \times RDCA$, 1.15 for $KZ \times RDCAP$, and 0.80 for $KZ \times RDA$.
- The standalone KZ coefficient is always tiny and insignificant, which is exactly the point: the constraints premium is mostly an *interaction* effect, not a sample-wide slope.
- When the paper adds $KZ \times R\&D \times \ln(ME)$, the two-way interaction stays positive and significant. So the result is not merely a relabeled size anomaly.

5.8 Table 4: Constrained-unconstrained returns across low- and high-R&D subsamples

Table 4
The returns of the constrained-unconstrained portfolios across subsamples split by R&D

	Panel A: KZ Index		Panel B: VW Index		Panel C: SA Index		Panel D: Size		Panel E: Age	
	Raw return	Industry-adjusted return	Raw return	Industry-adjusted return	Raw return	Industry-adjusted return	Raw return	Industry-adjusted return	Raw return	Industry-adjusted return
Low RDCA	0.16 (0.76)	0.16 (0.97)	-0.01 (-0.05)	0.11 (0.37)	0.11 (0.59)	-0.17 (-0.54)	-0.09 (-0.17)	0.72 (2.70)	0.41 (2.85)	-0.26 (-1.32)
const.-unc.	0.80 (2.00)	0.47 (2.16)	0.48 (1.43)	0.55 (2.73)	0.54 (1.36)	0.47 (1.33)	0.48 (1.42)	1.22 (6.78)	0.72 (4.06)	0.63 (1.07)
Low RDS	0.18 (0.86)	0.17 (1.01)	0.06 (0.42)	0.05 (0.18)	0.14 (0.45)	-0.20 (-0.60)	-0.14 (-0.40)	0.86 (3.31)	0.55 (3.82)	-0.23 (-1.16)
const.-unc.	0.80 (2.00)	0.47 (2.16)	0.48 (1.43)	0.55 (2.73)	0.54 (1.36)	0.47 (1.33)	0.48 (1.42)	1.22 (6.78)	0.72 (4.06)	0.63 (1.07)
Low RDE	0.18 (0.86)	0.17 (1.01)	0.06 (0.42)	0.05 (0.18)	0.14 (0.45)	-0.20 (-0.60)	-0.14 (-0.40)	0.86 (3.31)	0.55 (3.82)	-0.23 (-1.16)
const.-unc.	0.80 (2.00)	0.47 (2.16)	0.48 (1.43)	0.55 (2.73)	0.54 (1.36)	0.47 (1.33)	0.48 (1.42)	1.22 (6.78)	0.72 (4.06)	0.63 (1.07)
Low RDCA	0.16 (0.76)	0.16 (0.97)	-0.01 (-0.05)	0.11 (0.37)	0.11 (0.59)	-0.17 (-0.54)	-0.09 (-0.17)	0.72 (2.70)	0.41 (2.85)	-0.26 (-1.32)
const.-unc.	0.80 (2.00)	0.47 (2.16)	0.48 (1.43)	0.55 (2.73)	0.54 (1.36)	0.47 (1.33)	0.48 (1.42)	1.22 (6.78)	0.72 (4.06)	0.63 (1.07)
Low RDS	0.18 (0.86)	0.17 (1.01)	0.06 (0.42)	0.05 (0.18)	0.14 (0.45)	-0.20 (-0.60)	-0.14 (-0.40)	0.86 (3.31)	0.55 (3.82)	-0.23 (-1.16)
const.-unc.	0.80 (2.00)	0.47 (2.16)	0.48 (1.43)	0.55 (2.73)	0.54 (1.36)	0.47 (1.33)	0.48 (1.42)	1.22 (6.78)	0.72 (4.06)	0.63 (1.07)
Low RDE	0.18 (0.86)	0.17 (1.01)	0.06 (0.42)	0.05 (0.18)	0.14 (0.45)	-0.20 (-0.60)	-0.14 (-0.40)	0.86 (3.31)	0.55 (3.82)	-0.23 (-1.16)
const.-unc.	0.80 (2.00)	0.47 (2.16)	0.48 (1.43)	0.55 (2.73)	0.54 (1.36)	0.47 (1.33)	0.48 (1.42)	1.22 (6.78)	0.72 (4.06)	0.63 (1.07)
Low RDCA	0.16 (0.76)	0.16 (0.97)	-0.01 (-0.05)	0.11 (0.37)	0.11 (0.59)	-0.17 (-0.54)	-0.09 (-0.17)	0.72 (2.70)	0.41 (2.85)	-0.26 (-1.32)
const.-unc.	0.80 (2.00)	0.47 (2.16)	0.48 (1.43)	0.55 (2.73)	0.54 (1.36)	0.47 (1.33)	0.48 (1.42)	1.22 (6.78)	0.72 (4.06)	0.63 (1.07)
Low RDS	0.18 (0.86)	0.17 (1.01)	0.06 (0.42)	0.05 (0.18)	0.14 (0.45)	-0.20 (-0.60)	-0.14 (-0.40)	0.86 (3.31)	0.55 (3.82)	-0.23 (-1.16)
const.-unc.	0.80 (2.00)	0.47 (2.16)	0.48 (1.43)	0.55 (2.73)	0.54 (1.36)	0.47 (1.33)	0.48 (1.42)	1.22 (6.78)	0.72 (4.06)	0.63 (1.07)
Low RDE	0.18 (0.86)	0.17 (1.01)	0.06 (0.42)	0.05 (0.18)	0.14 (0.45)	-0.20 (-0.60)	-0.14 (-0.40)	0.86 (3.31)	0.55 (3.82)	-0.23 (-1.16)
const.-unc.	0.80 (2.00)	0.47 (2.16)	0.48 (1.43)	0.55 (2.73)	0.54 (1.36)	0.47 (1.33)	0.48 (1.42)	1.22 (6.78)	0.72 (4.06)	0.63 (1.07)

Table 4
Continued

	Panel A: KZ Index		Panel B: VW Index		Panel C: SA Index		Panel D: Size		Panel E: Age	
	Raw return	Industry-adjusted return	Raw return	Industry-adjusted return	Raw return	Industry-adjusted return	Raw return	Industry-adjusted return	Raw return	Industry-adjusted return
Low RDCA	0.18 (0.96)	0.15 (0.91)	0.07 (0.44)	0.08 (0.29)	0.16 (0.86)	-0.23 (-0.80)	0.00 (-0.02)	-0.06 (-0.27)	0.35 (3.85)	-0.23 (-1.11)
const.-unc.	0.80 (2.00)	0.47 (2.16)	0.48 (1.43)	0.55 (2.73)	0.54 (1.36)	0.47 (1.33)	0.48 (1.42)	1.22 (6.78)	0.72 (4.06)	0.63 (1.07)
Low RDS	0.18 (0.96)	0.15 (0.91)	0.07 (0.44)	0.08 (0.29)	0.16 (0.86)	-0.23 (-0.80)	0.00 (-0.02)	-0.06 (-0.27)	0.35 (3.85)	-0.23 (-1.11)
const.-unc.	0.80 (2.00)	0.47 (2.16)	0.48 (1.43)	0.55 (2.73)	0.54 (1.36)	0.47 (1.33)	0.48 (1.42)	1.22 (6.78)	0.72 (4.06)	0.63 (1.07)
Low RDE	0.18 (0.96)	0.15 (0.91)	0.07 (0.44)	0.08 (0.29)	0.16 (0.86)	-0.23 (-0.80)	0.00 (-0.02)	-0.06 (-0.27)	0.35 (3.85)	-0.23 (-1.11)
const.-unc.	0.80 (2.00)	0.47 (2.16)	0.48 (1.43)	0.55 (2.73)	0.54 (1.36)	0.47 (1.33)	0.48 (1.42)	1.22 (6.78)	0.72 (4.06)	0.63 (1.07)
Low RDCA	0.16 (0.76)	0.16 (0.97)	-0.01 (-0.05)	0.11 (0.37)	0.11 (0.59)	-0.17 (-0.54)	-0.09 (-0.17)	0.72 (2.70)	0.41 (2.85)	-0.26 (-1.32)
const.-unc.	0.80 (2.00)	0.47 (2.16)	0.48 (1.43)	0.55 (2.73)	0.54 (1.36)	0.47 (1.33)	0.48 (1.42)	1.22 (6.78)	0.72 (4.06)	0.63 (1.07)
Low RDS	0.18 (0.86)	0.17 (1.01)	0.06 (0.42)	0.05 (0.18)	0.14 (0.45)	-0.20 (-0.60)	-0.14 (-0.40)	0.86 (3.31)	0.55 (3.82)	-0.23 (-1.16)
const.-unc.	0.80 (2.00)	0.47 (2.16)	0.48 (1.43)	0.55 (2.73)	0.54 (1.36)	0.47 (1.33)	0.48 (1.42)	1.22 (6.78)	0.72 (4.06)	0.63 (1.07)
Low RDE	0.18 (0.86)	0.17 (1.01)	0.06 (0.42)	0.05 (0.18)	0.14 (0.45)	-0.20 (-0.60)	-0.14 (-0.40)	0.86 (3.31)	0.55 (3.82)	-0.23 (-1.16)
const.-unc.	0.80 (2.00)	0.47 (2.16)	0.48 (1.43)	0.55 (2.73)	0.54 (1.36)	0.47 (1.33)	0.48 (1.42)	1.22 (6.78)	0.72 (4.06)	0.63 (1.07)

At the end of June of year t , I sort R&D-reporting firms with positive real sales growth into two R&D groups and three financial constraints groups independently. All sorting variables are for the fiscal year ending in year $t-1$ except size. The interaction of these groups forms six R&D-constraints portfolios, which are held for the next 12 months and reformed every year. I report the average value-weighted monthly return, characteristic-adjusted return by size, book-to-market, and momentum, and industry-adjusted return for the constrained-unconstrained portfolios created in the low and high R&D groups. Heteroscedasticity-robust t -statistics are reported in parentheses. $RDCA$ is R&D capital scaled by total assets. R&D capital is computed as the five-year cumulative R&D expenditures assuming an annual depreciation rate of 20%. RDS , $RDCAP$, RDE , RDE , and RDA are R&D expenditures scaled by sales, capital expenditures, year-end market equity, number of employees, and assets. The KZ index, the VW index, and the SA index are indices of financial constraints. Size is market capitalization at the end of June of year t . Age is the number of years a firm has been on Compustat with a non-missing stock price. The sample is from 1975 to 2007.

(continued)

Read:

- **KZ example.** Among high-RDCA firms, the high-minus-low KZ portfolio earns 0.60% raw, 0.47% characteristic-adjusted, and 0.48% industry-adjusted per month. Among low-RDCA firms, the corresponding returns are only 0.16%, 0.16%, and -0.01% .
- **WW example.** Among high-RDME firms, the high-minus-low WW portfolio earns 0.82% raw, 0.87% characteristic-adjusted, and 0.80% industry-adjusted per month. Among low-RDME firms, the corresponding returns are negative and insignificant.
- **SA, size, and age.** The same qualitative pattern shows up for the SA index, size, and age: the constraints premium is small in low-R&D firms and large in high-R&D firms.
- This is the paper's most direct evidence for the first headline claim.

5.9 Table 5: Triple sorts by R&D and size

Table 5
The returns of the constrained-minus-unconstrained portfolios across subsamples split by R&D and size

	Panel A: KZ Index			Panel B: WW Index			Panel C: SA Index			Panel D: Age		
	Raw return	Charact.-adj. return	Industry-adjusted return	Raw return	Charact.-adj. return	Industry-adjusted return	Raw return	Charact.-adj. return	Industry-adjusted return	Raw return	Charact.-adj. return	Industry-adjusted return
Low RDCA, small, const.-uncost.	-0.10 (-0.47)	-0.09 (-0.42)	-0.16 (-0.76)	-0.12 (-0.35)	0.33 (1.03)	-0.02 (-0.05)	-0.09 (-0.81)	-0.18 (-2.17)	-0.52 (-3.14)	-0.47 (-2.04)	-0.61 (-2.68)	-0.49 (-2.08)
Low RDCA, big, const.-uncost.	0.17 (0.81)	0.17 (0.89)	0.00 (-0.30)	-0.21 (-0.82)	-0.15 (-0.39)	-0.12 (-0.35)	-0.38 (-1.38)	-0.62 (-1.30)	-0.50 (-1.32)	-0.45 (-1.62)	-0.48 (-1.64)	-0.39 (-1.64)
High RDCA, small, const.-uncost.	0.30 (0.48)	0.15 (0.61)	0.15 (0.65)	-0.24 (-0.47)	-0.23 (-0.44)	0.28 (0.56)	-1.11 (-3.37)	-0.61 (-2.08)	-1.13 (-3.50)	-0.36 (-1.54)	-0.07 (-0.31)	-0.38 (-1.61)
High RDCA, big, const.-uncost.	0.39 (1.92)	0.45 (0.95)	0.47 (1.74)	-0.04 (-0.09)	-0.04 (-0.13)	-0.04 (-0.14)	0.32 (0.77)	-0.01 (-0.05)	0.22 (0.60)	0.09 (0.73)	0.21 (1.79)	0.00 (-0.15)
Low RDME, small, const.-uncost.	-0.14 (-0.73)	-0.12 (-0.65)	-0.15 (-0.78)	-0.02 (-0.07)	0.42 (1.35)	0.01 (0.04)	-0.67 (-2.74)	-0.21 (-0.98)	-0.55 (-2.37)	-0.79 (-3.32)	-0.57 (-2.38)	-0.67 (-2.95)
Low RDME, big, const.-uncost.	0.30 (0.91)	0.18 (1.03)	0.07 (0.45)	-0.14 (-0.69)	-0.12 (-0.69)	-0.12 (-0.69)	-0.92 (-2.53)	-0.12 (-0.53)	-0.91 (-2.45)	-0.36 (-1.50)	-0.04 (-0.39)	-0.33 (-0.94)
High RDME, small, const.-uncost.	-0.01 (-0.06)	0.01 (0.03)	0.02 (0.07)	-0.14 (-0.34)	-0.09 (-0.04)	-0.12 (-0.21)	-0.92 (-2.79)	-0.12 (-1.68)	-0.91 (-2.87)	-0.36 (-1.52)	-0.04 (-0.19)	-0.33 (-1.41)
High RDME, big, const.-uncost.	0.40 (2.02)	0.44 (1.90)	0.44 (1.65)	0.09 (0.22)	-0.03 (-0.12)	0.08 (0.15)	0.32 (0.82)	-0.01 (-0.03)	0.22 (0.70)	0.24 (1.48)	0.12 (0.46)	0.06 (1.54)

(continued)

Table 5
Continued

	Panel A: KZ Index			Panel B: WW Index			Panel C: SA Index			Panel D: Age		
	Raw return	Charact.-adj. return	Industry-adjusted return	Raw return	Charact.-adj. return	Industry-adjusted return	Raw return	Charact.-adj. return	Industry-adjusted return	Raw return	Charact.-adj. return	Industry-adjusted return
Low RDCA, small, const.-uncost.	-0.11 (-0.52)	-0.10 (-0.47)	-0.14 (-0.70)	-0.13 (-0.38)	0.25 (0.82)	-0.09 (-0.28)	-0.35 (-0.58)	-0.13 (-0.58)	-0.35 (-0.58)	-0.13 (-0.58)	-0.35 (-0.58)	-0.49 (-2.08)
Low RDCA, big, const.-uncost.	0.16 (1.62)	0.29 (1.75)	0.11 (0.69)	-0.42 (-1.11)	-0.31 (-0.84)	-0.33 (-1.01)	-0.51 (-1.27)	-0.34 (-1.01)	-0.51 (-1.27)	-0.34 (-1.01)	-0.51 (-1.27)	-0.42 (-1.79)
High RDCA, small, const.-uncost.	0.10 (0.45)	0.14 (0.64)	0.16 (0.65)	-0.55 (-1.96)	-0.50 (-1.80)	-0.47 (-1.80)	-1.02 (-3.19)	-0.47 (-1.89)	-1.02 (-3.19)	-0.47 (-1.89)	-1.02 (-3.19)	-0.50 (-2.08)
High RDCA, big, const.-uncost.	0.49 (2.20)	0.39 (1.53)	0.54 (1.92)	0.05 (0.10)	0.05 (0.00)	0.03 (0.07)	0.41 (1.01)	-0.12 (-0.42)	0.34 (0.82)	0.59 (1.43)	0.05 (0.19)	0.20 (1.67)

At the end of June of year t , I sort R&D-reporting firms with positive real sales growth into two R&D groups, two size groups, and three financial constraints groups independently. All sorting variables are for the fiscal year ending in year $t-1$ except size. The interaction of these groups forms twelve R&D-size-constraints portfolios, which are held for the next 12 months and informed every year. I report the average value-weighted monthly returns, characteristic-adjusted returns by size, book-to-market, and momentum, and industry-adjusted returns for the constrained-minus-unconstrained portfolios created in the four R&D-size groups. Heteroscedasticity-robust t -statistics are reported in parentheses. RDCA is R&D capital scaled by total assets. RDME is the five-year cumulative R&D expenditures, assuming an annual depreciation rate of 20%. RDCA and RDME are R&D expenditures scaled by sales and capital expenditures. The KZ index, the WW index, and the SA index are indices of financial constraints. Age is the number of years a firm has been on Compustat with a non-missing stock price. The sample is for 1975–2007.

Read:

- The paper asks whether the Table 4 results are driven only by very small firms. The answer is mostly no.
- Among *big* high-RDCA firms, the high-minus-low KZ portfolio earns 0.59% raw, 0.45% characteristic-adjusted, and 0.47% industry-adjusted per month. Among *small* high-RDCA firms, the corresponding returns are only 0.10%, 0.13%, and 0.15%.
- The same broad message holds for firm age. For WW and SA, the signal becomes weaker once size is conditioned on, which makes sense because those indices are themselves highly collinear with size.
- The takeaway is that the positive constraints premium among high-R&D firms is not a microcap artifact.

5.10 Table 6: Factor loadings of constrained-minus-unconstrained portfolios

Read:

- The hedge portfolios formed among high-R&D firms load positively on the market and especially on the size factor. For example, the high-minus-low KZ portfolio in the high-RDCA subsample loads about 0.33 on the market factor and 0.76 on *SMB*, versus only 0.07 and 0.08 in the low-RDCA subsample.
- Loadings on *HML* are generally negative, and loadings on momentum and liquidity are usually weak.

- This is useful because it shows *what kind* of systematic risk rises when constraints matter for R&D-heavy firms: primarily market and small-firm risk.

Table 6 The risk factor loadings of the constrained-minus-unconstrained portfolios across subsamples split by R&D

	KZ index										WW index										SA index										
	Mkt	SMB	HML	mom	Liq	Mkt	SMB	HML	mom	Liq	Mkt	SMB	HML	mom	Liq	Mkt	SMB	HML	mom	Liq	Mkt	SMB	HML	mom	Liq	Mkt	SMB	HML	mom	Liq	
Low RDCA	0.07	0.08	0.26	0.22	-0.07	0.13	1.23	0.00	-0.05	-0.04	0.17	0.98	-0.37	-0.10	-0.09	0.17	0.98	-0.37	-0.10	-0.09	0.17	0.98	-0.37	-0.10	-0.09	0.17	0.98	-0.37	-0.10	-0.09	
const-unconst	(1.17)	(0.84)	(2.54)	(3.02)	(-1.38)	(2.36)	(13.91)	(-0.05)	(-0.89)	(-0.77)	(2.31)	(7.83)	(-1.35)	(-1.18)	(-1.11)	0.17	0.98	-0.37	-0.10	-0.09	0.17	0.98	-0.37	-0.10	-0.09	0.17	0.98	-0.37	-0.10	-0.09	
High RDCA	0.33	0.76	-0.39	0.17	-0.08	0.23	1.62	-0.34	0.01	0.06	0.29	1.60	-0.39	0.00	0.09	0.29	1.60	-0.39	0.00	0.09	0.29	1.60	-0.39	0.00	0.09	0.29	1.60	-0.39	0.00	0.09	
const-unconst	(2.10)	(7.82)	(-2.42)	(2.16)	(-1.37)	(4.03)	(17.13)	(-1.54)	(0.18)	(1.02)	(5.00)	(18.10)	(-2.23)	(-0.06)	(1.57)	0.29	1.60	-0.39	0.00	0.09	0.29	1.60	-0.39	0.00	0.09	0.29	1.60	-0.39	0.00	0.09	
Low RDE	0.14	0.15	-0.06	0.12	0.02	0.14	1.19	0.13	-0.13	-0.07	0.16	1.07	0.05	-0.08	-0.09	0.16	1.07	0.05	-0.08	-0.09	0.16	1.07	0.05	-0.08	-0.09	0.16	1.07	0.05	-0.08	-0.09	
const-unconst	(2.21)	(1.46)	(-0.50)	(1.58)	(0.29)	(2.39)	(13.72)	(1.45)	(-1.14)	(-1.06)	(2.56)	(10.15)	(0.55)	(-1.28)	(-1.22)	0.16	1.07	0.05	-0.08	-0.09	0.16	1.07	0.05	-0.08	-0.09	0.16	1.07	0.05	-0.08	-0.09	
High RDE	0.29	0.78	-0.19	0.11	-0.06	0.22	1.59	-0.33	0.05	0.05	0.29	1.45	-0.38	0.00	0.03	0.29	1.45	-0.38	0.00	0.03	0.29	1.45	-0.38	0.00	0.03	0.29	1.45	-0.38	0.00	0.03	
const-unconst	(4.26)	(8.49)	(-1.53)	(1.46)	(-0.94)	(3.81)	(15.80)	(-1.48)	(0.66)	(0.87)	(4.93)	(17.17)	(-4.12)	(0.02)	(0.66)	0.22	1.59	-0.33	0.05	0.05	0.22	1.59	-0.33	0.05	0.05	0.22	1.59	-0.33	0.05	0.05	
Low RDCA	0.15	0.25	-0.09	0.11	0.01	0.10	1.20	-0.11	-0.03	-0.10	0.12	1.08	-0.27	-0.14	-0.07	0.10	1.20	-0.11	-0.03	-0.10	0.12	1.08	-0.27	-0.14	-0.07	0.10	1.20	-0.11	-0.03	-0.10	-0.07
const-unconst	(2.50)	(2.35)	(-0.81)	(1.23)	(0.72)	(1.80)	(12.43)	(-1.23)	(-0.47)	(-1.68)	(1.72)	(8.54)	(-1.34)	(-1.60)	(-0.91)	0.10	1.20	-0.11	-0.03	-0.10	0.12	1.08	-0.27	-0.14	-0.07	0.10	1.20	-0.11	-0.03	-0.10	-0.07
High RDCA	0.29	0.89	-0.13	0.23	-0.16	0.26	1.63	-0.18	0.00	0.08	0.31	1.58	-0.29	0.04	0.09	0.26	1.63	-0.18	0.00	0.08	0.31	1.58	-0.29	0.04	0.09	0.26	1.63	-0.18	0.00	0.08	0.31
const-unconst	(4.16)	(9.68)	(-1.10)	(2.80)	(-2.73)	(4.16)	(14.80)	(-1.76)	(0.00)	(1.24)	(4.82)	(17.93)	(-2.89)	(0.31)	(1.40)	0.26	1.63	-0.18	0.00	0.08	0.31	1.58	-0.29	0.04	0.09	0.26	1.63	-0.18	0.00	0.08	0.31
Low RDE	0.18	0.23	0.24	0.12	-0.02	0.18	1.27	-0.08	-0.13	-0.11	0.26	1.34	-0.39	-0.11	0.02	0.18	1.27	-0.08	-0.13	-0.11	0.26	1.34	-0.39	-0.11	0.02	0.18	1.27	-0.08	-0.13	-0.11	0.02
const-unconst	(1.52)	(2.18)	(2.25)	(1.40)	(-0.40)	(3.17)	(14.73)	(-0.89)	(-1.29)	(-1.80)	(4.30)	(13.50)	(-1.83)	(-1.68)	(0.39)	0.18	1.27	-0.08	-0.13	-0.11	0.26	1.34	-0.39	-0.11	0.02	0.18	1.27	-0.08	-0.13	-0.11	0.02
High RDE	0.28	0.60	0.07	0.17	-0.14	0.14	1.61	-0.57	0.02	0.04	0.15	1.42	-0.62	0.03	0.05	0.28	0.60	0.07	0.17	-0.14	0.14	1.61	-0.57	0.02	0.04	0.15	1.42	-0.62	0.03	0.05	0.28
const-unconst	(3.94)	(4.26)	(0.55)	(1.89)	(-1.65)	(2.77)	(14.56)	(-1.51)	(0.26)	(0.58)	(2.51)	(19.53)	(-6.14)	(0.55)	(0.92)	0.15	1.42	-0.62	0.03	0.05	0.15	1.42	-0.62	0.03	0.05	0.15	1.42	-0.62	0.03	0.05	0.15
Low RDCA	0.11	0.18	-0.05	0.11	0.00	0.17	1.11	0.03	-0.07	-0.05	0.19	0.98	-0.12	-0.05	-0.04	0.11	0.18	-0.05	0.11	0.00	0.17	1.11	0.03	-0.07	-0.05	0.19	0.98	-0.12	-0.05	-0.04	0.11
const-unconst	(1.76)	(1.93)	(-0.47)	(1.57)	(0.04)	(3.00)	(13.20)	(0.30)	(-1.12)	(-0.80)	(3.09)	(9.00)	(-1.23)	(-0.83)	(-0.55)	0.17	1.11	0.03	-0.07	-0.05	0.19	0.98	-0.12	-0.05	-0.04	0.19	0.98	-0.12	-0.05	-0.04	0.19
High RDCA	0.33	0.65	-0.08	0.14	0.00	0.19	1.66	-0.37	-0.03	0.07	0.26	1.52	-0.43	-0.07	0.03	0.33	0.65	-0.08	0.14	0.00	0.19	1.66	-0.37	-0.03	0.07	0.26	1.52	-0.43	-0.07	0.03	0.33
const-unconst	(4.95)	(8.24)	(-0.63)	(1.54)	(0.40)	(3.31)	(15.89)	(-1.34)	(-0.48)	(0.99)	(4.44)	(17.60)	(-4.79)	(-1.08)	(0.52)	0.19	1.66	-0.37	-0.03	0.07	0.26	1.52	-0.43	-0.07	0.03	0.26	1.52	-0.43	-0.07	0.03	0.26

At the end of June 2007, we split R&D-reporting firms with positive real sales growth into two R&D groups and three financial constraints groups (unconstrained), the intersection of the R&D and financial constraints groups. We then split the unconstrained group into two R&D groups and three financial constraints groups. The intersection of the R&D and financial constraints groups is R&D capital spend by total assets. R&D capital is the five-year cumulative R&D expenditures, assuming an annual depreciation rate of 20%. RDE, RDCA, HML, mom, and Liq are the market, size, book-to-market, momentum, and liquidity factors, respectively. Mkt, SMB, HML, mom, and Liq are the market, size, book-to-market, momentum, and liquidity factors. The sample is from 1975 to 2007.

At the end of June of year t , I sort R&D-reporting firms with positive real sales growth into two R&D groups and three financial constraints groups independently. The intersection of these groups forms six R&D-constraints portfolios, which are held for the next 12 months and reformed every year. I also form a constrained-minus-unconstrained portfolio in each R&D group. RDCA is R&D capital scaled by total assets. R&D capital is the five-year cumulative R&D expenditures, assuming an annual depreciation rate of 20%. RDCA, RDCA, RDCA, RDCA, RDCA, and RDCA are R&D capital scaled by sales, capital expenditure, year-end market equity, number of employees, and assets. The KZ index, the WW index, and the SA index are indices of financial constraints. All sorting variables are for the fiscal year ending in year $t-1$. I report the value-weighted monthly excess returns of the constrained-minus-unconstrained portfolios on the market, size, book-to-market, momentum, and liquidity factors. I report the slopes on these factors and the heteroscedasticity-robust t -statistics in parentheses. Mkt, SMB, HML, Mom, and Liq are the market, size, book-to-market, momentum, and liquidity factors. The sample is from 1973 to 2007.

5.11 Table 7: High-minus-low R&D returns across financial-constraints subsamples

Read:

- The positive R&D premium appears only when firms are financially constrained. Among small firms, the high-minus-low RDCA portfolio earns 0.69% raw, 0.86% characteristic-adjusted, and 0.69% industry-adjusted per month; among big firms the same spread is essentially zero.
- The same pattern appears for age. Among young firms, the high-minus-low RDCA spread is 0.94% raw and 0.71% characteristic-adjusted; among old firms it is close to zero.
- Using the SA index, the high-minus-low RDS portfolio earns 0.59%, 0.60%, and 0.52% per month in the high-SA group but essentially nothing in the low-SA group.
- This table is the mirror image of Table 4 and the strongest evidence for the second headline claim.

5.12 Table 8: Factor loadings of high-minus-low R&D portfolios

Read:

Table 8
The risk factor loadings of the high-minus-low R&D portfolios across subsamples split by financial constraints

	RDCA										RDCA										RDS										RDME									
	Mkt	SMB	HML	mom	Liq	Mkt	SMB	HML	mom	Liq	Mkt	SMB	HML	mom	Liq	Mkt	SMB	HML	mom	Liq	Mkt	SMB	HML	mom	Liq	Mkt	SMB	HML	mom	Liq	Mkt	SMB	HML	mom	Liq					
Big size	-0.08	0.11	-0.43	-0.09	0.06	-0.05	0.10	-0.62	-0.11	0.08	-0.04	0.17	-0.54	-0.16	0.11	0.02	0.82	0.48	0.22	-0.04	0.06	0.15	0.10	-0.43	-0.09	0.06	-0.05	0.10	-0.62	-0.11	0.08	-0.04	0.17	-0.54	-0.16	0.11	0.02			
H-L R&D	(-1.74)	(1.43)	(-5.43)	(-1.29)	(1.51)	(-1.26)	(1.32)	(-6.10)	(-1.67)	(2.62)	(-0.81)	(2.10)	(-6.66)	(-2.23)	(3.90)	(0.29)	(3.11)	(2.17)	(-0.46)	(1.11)	0.06	0.09	0.60	-0.53	-0.07	0.12	0.11	0.59	-0.53	-0.08	0.18	0.14	0.89	-0.63	-0.08	0.11	0.11			
Small size	0.09	0.60	-0.53	-0.07	0.12	0.11	0.59	-0.53	-0.08	0.18	0.14	0.89	-0.63	-0.08	0.11	0.11	0.56	-0.32	-0.02	0.15	0.06	0.09	0.60	-0.53	-0.07	0.12	0.11	0.59	-0.53	-0.08	0.18	0.14	0.89	-0.63	-0.08	0.11	0.11			
H-L R&D	(1.65)	(5.83)	(-4.20)	(-1.17)	(1.80)	(2.09)	(4.86)	(-5.58)	(-1.20)	(2.43)	(2.15)	(6.45)	(-6.65)	(-1.15)	(1.77)	(1.80)	(4.91)	(-3.68)	(-0.29)	(2.22)	0.06	0.09	0.60	-0.53	-0.07	0.12	0.11	0.59	-0.53	-0.08	0.18	0.14	0.89	-0.63	-0.08	0.11	0.11			
Old age	-0.08	0.10	-0.41	-0.11	0.06	-0.09	0.05	-0.53	-0.15	0.05	-0.08	0.11	-0.41	-0.13	0.07	0.05	0.48	0.18	-0.08	0.09	0.15	0.10	-0.43	-0.09	0.06	-0.05	0.10	-0.62	-0.11	0.08	-0.04	0.17	-0.54	-0.16	0.11	0.02				
H-L R&D	(-1.73)	(1.25)	(-4.81)	(-1.67)	(1.70)	(-1.95)	(0.71)	(-7.00)	(-2.70)	(1.58)	(-1.60)	(1.41)	(-4.91)	(-1.99)	(1.88)	(0.96)	(4.99)	(1.96)	(-0.96)	(1.60)	0.06	0.09	0.60	-0.53	-0.07	0.12	0.11	0.59	-0.53	-0.08	0.18	0.14	0.89	-0.63	-0.08	0.11	0.11			
Young age	0.13	0.52	-0.64	-0.12	0.06	0.16	0.31	-0.97	-0.03	0.06	0.21	0.31	-1.20	-0.15	0.10	-0.07	0.56	0.10	0.12	-0.08	0.15	0.10	-0.43	-0.09	0.06	-0.05	0.10	-0.62	-0.11	0.08	-0.04	0.17	-0.54	-0.16	0.11	0.02				
H-L R&D	(1.60)	(2.79)	(-4.87)	(-1.32)	(0.47)	(2.01)	(2.59)	(-7.09)	(-0.28)	(0.87)	(2.39)	(2.86)	(-9.31)	(-1.39)	(1.48)	(-0.89)	(5.05)	(0.74)	(1.48)	(-1.06)	0.06	0.09	0.60	-0.53	-0.07	0.12	0.11	0.59	-0.53	-0.08	0.18	0.14	0.89	-0.63	-0.08	0.11	0.11			
Low SA	-0.08	0.07	-0.43	-0.10	0.06	-0.07	0.04	-0.57	-0.13	0.05	-0.06	0.10	-0.47	-0.16	0.08	0.02	0.46	0.21	-0.04	0.05	0.15	0.10	-0.43	-0.09	0.06	-0.05	0.10	-0.62	-0.11	0.08	-0.04	0.17	-0.54	-0.16	0.11	0.02				
H-L R&D	(-1.70)	(0.99)	(-5.24)	(-1.37)	(1.64)	(-1.67)	(0.64)	(-7.50)	(-2.70)	(1.58)	(-1.60)	(1.41)	(-4.91)	(-1.99)	(1.88)	(0.96)	(4.99)	(1.96)	(-0.96)	(1.60)	0.06	0.09	0.60	-0.53	-0.07	0.12	0.11	0.59	-0.53	-0.08	0.18	0.14	0.89	-0.63	-0.08	0.11	0.11			
High SA	-0.09	0.46	-0.34	-0.01	0.28	0.18	0.70	-0.37	-0.09	0.42	0.26	0.10	-0.70	-0.42	0.24	0.26	0.10	0.70	-0.42	0.24	0.26	0.10	-0.70	-0.42	0.24	0.26	0.10	-0.70	-0.42	0.24	0.26	0.10	-0.70	-0.42	0.24	0.26	0.10			
H-L R&D	(-1.66)	(4.04)	(-3.34)	(-0.01)	(2.38)	(1.58)	(3.70)	(-3.78)	(-0.93)	(4.24)	(2.56)	(1.10)	(-3.70)	(-4.82)	(2.44)	(2.64)	(1.37)	(4.67)	(-4.19)	(1.41)	0.06	0.09	0.60	-0.53	-0.07	0.12	0.11	0.59	-0.53	-0.08	0.18	0.14	0.89	-0.63	-0.08	0.11	0.11			
Low KZ	-0.16	-0.22	0.31	0.12	-0.01	-0.09	-0.09	-0.02	0.08	-0.10	-0.09	-0.52	-0.01	-0.09	-0.06	0.41	-0.09	-0.06	0.41	-0.09	-0.06	0.41	-0.09	-0.06	0.41	-0.09	-0.06	0.41	-0.09	-0.06	0.41	-0.09	-0.06	0.41	-0.09	-0.06	0.41			
H-L R&D	(-1.62)	(-2.22)	(3.11)	(1.21)	(-0.01)	(-0.92)	(-0.92)	(0.08)	(-1.10)	(-1.09)	(-0.52)	(-0.01)	(-0.92)	(-0.60)	(3.90)	(-0.92)	(-0.60)	(3.90)	(-0.92)	(-0.60)	(3.90)	(-0.92)	(-0.60)	(3.90)	(-0.92)	(-0.60)	(3.90)	(-0.92)	(-0.60)	(3.90)	(-0.92)	(-0.60)	(3.90)	(-0.92)	(-0.60)	(3.90)	(-0.92)	(-0.60)	(3.90)	
High KZ	0.09	0.66	-0.77	0.08	-0.04	0.03	0.63	-0.63	-0.10	-0.09	0.03	0.54	-0.66	-0.02	0.05	0.67	-0.01	0.05	0.67	-0.01	0.05	0.67	-0.01	0.05	0.67	-0.01	0.05	0.67	-0.01	0.05	0.67	-0.01	0.05	0.67	-0.01	0.05	0.67			
H-L R&D	(1.60)	(6.66)	(-7.97)	(0.82)	(-0.40)	(0.31)	(6.30)	(-6.63)	(-1.10)	(-1.09)	(0.31)	(5.54)	(-6.66)	(-0.22)	(0.65)	(6.67)	(-0.01)	(0.65)	(6.67)	(-0.01)	(0.65)	(6.67)	(-0.01)	(0.65)	(6.67)	(-0.01)	(0.65)	(6.67)	(-0.01)	(0.65)	(6.67)	(-0.01)	(0.65)	(6.67)	(-0.01)	(0.65)	(6.67)			
Low WIND	-0.11	0.38	-0.38	-0.09	0.04	-0.07	-0.58	-0.11	0.06	-0.06	0.04	-0.48	-0.16	0.09	0.03	0.43	0.24	-0.06	0.03	0.43	0.24	-0.06	0.03	0.43	0.24	-0.06	0.03	0.43	0.24	-0.06	0.03	0.43	0.24	-0.06	0.03	0.43	0.24	-0.06	0.03	
H-L R&D	(-1.17)	(1.44)	(-4.77)	(-1.28)	(1.17)	(-1.47)	(-5.80)	(-1.66)	(1.40)	(-0.60)	(0.40)	(-4.80)	(-1.66)	(1.40)	(-0.60)	(0.40)	(4.30)	(2.63)	(0.43)	(2.63)	(0.43)	(2.63)	(0.43)	(2.63)	(0.43)	(2.63)	(0.43)	(2.63)	(0.43)	(2.63)	(0.43)	(2.63)	(0.43)	(2.63)	(0.43)	(2.63)	(0.43)			
High WIND	0.20	0.35	-0.35	-0.28	0.19	0.24	0.32	-0.31	-0.11	0.20	0.25	0.36	-0.32	(1.11)	0.20	0.25	0.36	-0.32	(1.11)	0.20	0.25	0.36	-0.32	(1.11)	0.20	0.25	0.36	-0.32	(1.11)	0.20	0.25	0.36	-0.32	(1.11)	0.20	0.25	0.36	-0.32	(1.11)	
H-L R&D	(0.94)	(3.57)	(-3.53)	(-2.28)	(1.79)	(2.34)	(3.25)	(-3.11)	(-1.16)	(2.80)	(3.25)	0.36	(-3.28)	(1.11)	0.20	0.25	0.36	(-3.28)	(1.11)	0.20	0.25	0.36	(-3.28)	(1.11)	0.20	0.25	0.36	(-3.28)	(1.11)	0.20	0.25	0.36	(-3.28)	(1.11)	0.20	0.25	0.36	(-3.28)	(1.11)	

As at the end of June of year 1, I sort R&D portfolios with positive real sales growth into two R&D groups and three financial constraints groups independently. The intersection of the two R&D groups and three financial constraints groups is used to form the high-minus-low R&D portfolios. The high-minus-low R&D portfolios are used to form the high-minus-low R&D portfolios. The high-minus-low R&D portfolios are used to form the high-minus-low R&D portfolios. The high-minus-low R&D portfolios are used to form the high-minus-low R&D portfolios. The high-minus-low R&D portfolios are used to form the high-minus-low R&D portfolios. The high-minus-low R&D portfolios are used to form the high-minus-low R&D portfolios. The high-minus-low R&D portfolios are used to form the high-minus-low R&D portfolios. The high-minus-low R&D portfolios are used to form the high-minus-low R&D portfolios. The high-minus-low R&D portfolios are used to form the high-minus-low R&D portfolios. 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The high-minus-low R&D portfolios are used to form the high-minus-low R&D portfolios. The high-minus-low R&D portfolios are used to form the high-minus-low R&D portfolios. The high-minus-low R&D portfolios are used to form the high-minus-low R&D portfolios. The high-minus-low R&D portfolios are used to form the high-minus-low R&D portfolios. The high-minus-low R&D portfolios are used to form the high-minus-low R&D portfolios. The high-minus-low R&D portfolios are used to form the high-minus-low R&D portfolios. The

- When the high-minus-low R&D hedge is formed within constrained firms, it loads much more strongly on *SMB* than when it is formed within unconstrained firms.
- The constrained high-R&D hedge also sometimes loads more on the market and liquidity factors. For example, in the high-SA subsample, the high-minus-low RDME portfolio has much stronger systematic exposures than in the low-SA subsample.
- The economic message is that high R&D becomes risky mainly when it is combined with financing frictions. The return premium is not just compensating investors for “innovation” in the abstract.

6 Short summary

- **Conclusion.** Financial constraints and R&D are jointly priced. Each characteristic matters most when the other is high.
- **Intuition.** Because R&D is hard to scale down smoothly, constrained firms face a suspension margin that makes cash-flow shocks more dangerous and more systematic.
- **Empirical lesson.** Many seemingly mixed anomaly results become much sharper once the relevant interaction variable is added. The average slope can look weak even when the conditional slope is strong.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Main finding.** The paper establishes a robust positive interaction between financial constraints and R&D intensity in the cross-section of expected stock returns.
- **What happens to the constraints premium?** It is concentrated among R&D-intensive firms.
- **What happens to the R&D premium?** It is concentrated among financially constrained firms.
- **How credible is it?** The pattern survives multiple constraints measures, multiple R&D scalings, Fama–MacBeth regressions, double sorts, triple sorts, and adjusted-return calculations.
- **Mechanism evidence.** The factor-loading results point mainly to higher market and small-firm systematic exposure when firms are both constrained and R&D-intensive.
- **What the paper clarifies.** Mixed evidence on the financial-constraints–return relation in prior work is likely due to omitting R&D intensity, while the positive R&D–return relation may itself be driven by financing constraints.

7.2 Economic intuition

- R&D spending is lumpy, technologically disciplined, and hard to reverse without destroying option value.

- A financing shortfall therefore has a first-order effect on project continuation, unlike ordinary physical investment where firms can often adjust gradually.
- Investors demand a premium when they hold firms whose future value is highly exposed to that continuation margin. That exposure is strongest when the firm both needs a lot of R&D spending and has weak access to finance.

7.3 Takeaway

This is a very clean interaction paper. The real object being priced is not “financial constraints” alone and not “R&D” alone, but the combination of high required intangible investment and low external financing ability. Once the paper focuses on that joint margin, both the theory and the data line up: constrained R&D firms earn higher expected returns because they are closer to suspension, and suspension makes systematic cash-flow risk more painful.